



ACL Rehabilitation: How Can We Lessen Injury Rates?

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Rehabilitation is commonly used following anterior cruciate ligament (ACL) injury, though specific protocols are widely varied. Regardless of whether the injury is managed operatively or nonoperatively, quadriceps strengthening and return to full range of motion are the main focus points of rehabilitation, both to regain function and prevent future injury. Preoperative ACL rehabilitation has been shown to improve clinical outcomes. Home exercise programs can also provide good clinical outcomes when compared to traditional in-person physical therapy. A multitude of specific modalities (bracing, functional testing, neuromuscular training, electrostimulation, cryotherapy, and blood flow restriction therapy) may also provide benefit in ACL rehabilitation. Analysis of biomechanics during rehabilitation allows for customization of rehabilitation approaches, provides insight on muscular deficiencies, and helps ACL-injured patients re-establish functionality with the goal of preventing future ipsilateral or contralateral injury. Given the lack of consensus on rehabilitation protocols following ACL injury, the purpose of this review was to examine the efficacy of a variety of protocols and specific modalities and discuss their roles in the prevention of future ACL injury.

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Introduction

Anterior cruciate ligament (ACL) injuries can be devastating in athletes, highlighting the importance of adequate rehabilitation methods both to prevent injury and to decrease re-tear rate once return to sport occurs. Previous studies have determined the incidence of ACL rupture to be 36–69 per 100,000 person-years.^{1–3} The incidence is higher in females and peaks at a younger age range (ages 14–18) compared to males (ages 19–25).^{1,4}

Following ACL reconstruction (ACLR), re-tear continues to be a challenging problem to manage, especially amongst young athletes. Recent studies have found the rate of graft rupture to be 2.8%–23.5% following ACLRs.^{5–7} Risk factors for ipsilateral graft failure are younger age and higher activity

level.^{8–10} The rate of ACL rupture in the contralateral knee is 12%, which may be due to less rehabilitation of the contralateral knee following the initial injury, failure to correct deficiencies in the patient's kinetic chain, proprioceptive system, or the contralateral knee acting as a form of protection for the initially injured knee.^{11,12} Regardless of graft or fixation type, which are beyond the scope of this review, rehabilitation is a necessary component of returning to normal function after an ACL injury. Physical therapy is crucial to return an athlete to normal function and strength, as well as to improve mechanics to prevent injury to the surgical and contralateral extremity. The purpose of this review article was to examine the efficacy of various rehabilitation methods following ACL injury.

ACL Rehabilitation

Nonoperative Management

Although ligament reconstruction is typically used to manage acute ACL injury, nonoperative management is also an

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option. In a cohort study evaluating the association of early copers classification with 2-year success in patients treated with ACLR or nonoperatively, Thoma et al found that non-operative management can lead to good outcomes if patients are identified early after ACL injury as potential copers (athletes with dynamic stability and good patient-reported outcomes) rather than non-copers (athletes with recurrent dynamic instability episodes despite progressive rehabilitation).¹³ Patients in either group were able to self-select whether to undergo surgical management over the course of pre-operative rehabilitation. At 2-year follow-up, potential copers in the ACLR and nonoperative groups were more successful (2.7 and 2.9 times the odds of success, respectively) than ACLR non-copers, suggesting the impact of early copers classification on outcome regardless of management type.¹³

Paterno outlined a nonoperative course of ACL rehabilitation with a focus on addressing post-acute injury impairments, maximizing strength, and ensuring that activities can be performed without instability episodes prior to return to sport. In the acute period, hemarthrosis should be aggressively managed, cryotherapy and compression should be used to decrease effusion, and neuromuscular electric stimulation (NMES) should be considered if a patient lacks a strong quadriceps muscle contraction (Fig. 1).¹⁴ During this early rehabilitation phase, the main goal for the patient should be return to full range of motion. Initiation of quadriceps strengthening exercises (open kinetic chain [OKC] to address isolated quadriceps weakness¹⁵ and closed kinetic chain [CKC] to strengthen the quadriceps muscle with re-incorporation into dynamic movement) should occur at this time in addition to strengthening exercises for the hamstring, hip, and core muscles. Knee extension with OKC should be limited from 30-100 degrees of flexion.¹⁴ Following achievement of full range of motion, effusion resolution, and improvement in strength, Paterno suggests the addition of balance, proprioception, cardiovascular conditioning, and neuromuscular interventions to the rehabilitation protocol. The final phase of rehabilitation is focused on agility and sports-specific training.¹⁴

Preoperative Rehabilitation

Preoperative rehabilitation, also known as prehabilitation, is intended to prepare patients undergoing ACLR for immobility following surgery and to improve motion prior to the procedure. In a systematic review, Carter et al identified 3 studies (122 total patients) assessing postoperative outcomes in patients that went through prehabilitation and found variation throughout prehabilitation protocols. Program length ranged from 10 to 24 sessions and lacked consensus on whether exercises were to be supervised or performed at home.¹⁶ The protocols were also varied in focus: neuromuscular exercises, strengthening and balance, or supervised resistance and balance training were all recommended. Carter et al concluded that evidence surrounding use of prehabilitation is limited and of low quality due to the high risk of bias and variability present in each study.¹⁶ Nevertheless, Kim et al concluded that a 4-week prehabilitation course led to longer single-leg hop distances and lower quadriceps strength deficits compared to patients that did not undergo prehabilitation.¹⁷ Shaarani et al also observed improved single-leg hop test scores in addition to higher Cincinnati Knee Rating scores 12 weeks postoperatively compared to patients who did not undergo prehabilitation.¹⁸ Moreover, Failla et al found that extended prehabilitation consisting of progressive strengthening and neuromuscular training led to better functional outcomes (assessed with Knee Injury and Osteoarthritis Outcome Scores (KOOS)) and higher return to sports rates compared standard prehabilitation when assessed 2 years after ACLR.¹⁹ While controversial, the authors' preferred technique is to prescribe prehabilitation for those patients with significant preoperative swelling or limited motion in the absence of any mechanical block diagnosed on advanced imaging (ie bucket handle meniscus tear).

Physical Therapy

Rehabilitation protocols vary widely following ACL injury and are largely dependent on the patient's desired level of



Figure 1 NMES in ACLR rehabilitation. Electrostimulation exercises. (A) Electrostimulation during an open kinetic chain straight leg raise, and (B) Electrostimulation during a closed kinetic chain terminal knee extension with a band. (Color version of figure is available online.)

sport, return to play timeline, and presence of concomitant injuries. Following a systematic review of the literature, van Grinsven et al developed an evidence-based rehabilitation protocol that was accelerated, did not include postoperative bracing, and emphasized reduction of pain, swelling, inflammation.²⁰ Gupta et al highlighted the importance of core stabilization postoperatively, showing that the addition of core stability exercises in rehabilitation led to higher International Knee Documentation Committee (IKDC) scores, wider single and triple hop jump distances, and faster return to sports rates.²¹ In a systematic review of the literature, Makhni et al demonstrated inconsistency in rehabilitation protocols published online by academic orthopaedic programs. They found that of 42 rehabilitation protocols, 31% recommended preoperative rehabilitation, 14% specified range of motion exercises, and 19% suggested quadriceps strengthening exercises.²² Similar to rehabilitation in nonoperative therapy, immediate motion was recommended postoperatively by all 42 protocols reviewed, though some variability existed in when milestones should be reached. Strengthening exercises present in over 50% of the included protocols were straight-leg raise, quad sets and/or isometrics, mini-squats, hamstring curls, leg press, step up and/or down, and lunges. Proprioception exercises present in over 50% of included protocols were 1-leg balance and tilt and/or wobble and/or balance board (Fig. 2).²²



Figure 2 Proprioception exercise on balance board. (Color version of figure is available online.)

Table 1 Peebles et al ACLR Rehabilitation Protocol.

Timeframe	Rehabilitation Focus
Wk 0-6	Range of motion and muscle initiation
Wk 6-12	Muscular endurance base development
Wk 12-20	Muscular strength
Wk 20-32	Muscular power
Wk 32-54	Running progression, speed, and agility

For use in a young and active military population, Peebles et al outlined a 5 phase ACLR rehabilitation protocol (Table 1).²³ Phase progression is suggested if patients meet certain criteria, such as timeframe, degree of knee flexion, ambulation time, exercise endurance, and quadriceps strength index. The authors believe that to avoid issues with deconditioning, low intensity cardiovascular exercises should be initiated early in the recovery phase. In attempt to prevent compensation through the hip and ankle for decreased knee flexion, gait training should also be initiated early.²³

Quadriceps and hamstring muscles are the focus of muscular activation exercises during phase I, namely attaining quadriceps contraction in full extension and active heel slides or standing knee flexion. CKC exercises may be utilized in phase II (body weight squatting, single leg Romanian deadlift, and leg press). Though OKC exercises may be incorporated in phase III (muscular strength), CKC exercises should remain the focus.²³ There is clinical agreement on use of CKC exercises in ACL rehabilitation, but use of OKC is not as clearly defined.^{14,23,24} Fukuda et al assessed the impact of early (4 weeks postoperative) vs late (12 weeks postoperative) initiation of OKC exercises through evaluation of muscle strength, anterior knee laxity, and pain and function at postoperative follow-up time points: 12 weeks, 19 weeks, 25 weeks, and 17 months. Their findings revealed that early initiation led to improved quadriceps strength (determined by measuring maximum isometric voluntary contraction with a handheld dynamometer) at an earlier time point than late initiation - the early initiation group demonstrated improvement by 19 weeks follow-up while the late initiation group did not show improvement until 17 months follow-up. No difference existed between groups for anterior knee laxity, pain, function, or quadriceps strength at 17 month follow-up.²⁴

Home Exercise Programs

The success of home exercise programs is highly dependent on the level of patient motivation and the quality of patient education on how to perform each exercise. Home exercise programs, which are becoming increasingly popular due to the shift toward accelerated rehabilitation programs, have previously been shown to lead to similar subjective and objective outcomes as supervised rehabilitation.²⁵⁻²⁷ In a 2-4-year follow-up randomized clinical trial, Grant and Moh-tadi found no differences between patients treated with supervised or home-based rehabilitation in terms of range of motion, sagittal plane laxity, quadriceps strength ratio, or hamstring strength ratio. The home-based therapy group

went through 4 in-person physical therapy sessions within 3 months after surgery. Of note, minimal changes were observed between 3 months and 3 years postoperatively for each outcome measured except for quadriceps strength, which significantly improved over this time frame for both groups. Grant and Mohtadi also found that patients in the home-based therapy group had higher disease-specific quality of life scores 3 years postoperatively, although this difference did not meet the *a priori* definition of clinical relevance.²⁸

Through analysis of isokinetic knee strength and proprioception following ACLR, a recent randomized clinical trial by Lim et al found good outcomes for both supervised and home rehabilitation. No difference was observed between the groups for improvement in isokinetic knee strength compared to preoperative levels with the exception of no improvement in knee extension total work in the home-based group. The supervised rehabilitation group showed significant increases in proprioception scores compared to preoperative levels while the home-based rehabilitation group did not, which the authors attribute to the lack of feedback patients receive in home-based programs on posture and exercise position.²⁹

Specific Modalities and Emerging Trends

Accelerated rehabilitation programs have grown in popularity in an effort to both decrease cost of physical therapy incurred by patients and insurance companies and to return patients to activity sooner.²⁸ After a systematic review of the literature, van Grinsven et al concluded that accelerated rehabilitation (17-26 weeks) following ACLR leads to similar outcomes as standard programs (32-34 weeks) with respect to clinical, functional, and patient-reported outcomes.²⁰ However, these results should be considered within the realm of return to sport rates as this is the ultimate goal of ACLR. Arderm et al found that although return to any type of sport after ACLR was 81% in the general population, only 55% returned to competitive level.³⁰ Moreover, one systematic review found that 47% of patients that did not return to preinjury levels of sport within 1 year of ACLR had intentions of return to competitive play.³¹ A recent prospective cohort study also found that when young (15-30 years old) athletes returned to sport within 9 months of ACLR, there was a 7-fold increased risk of sustaining a second ACL injury compared to return to sport more than 9 months after surgery.³² These findings pose the question of whether postoperative rehabilitation protocols should be longer or more sport specialized to allow for full recovery and safe return to sport. Whether the program is accelerated or standard, patient motivation, and high-quality patient education are necessary for success.

Functional Testing

Functional testing (isokinetic strength, single limb hop, single leg squat, limb symmetry index (LSI)) is frequently used to assess progress in ACL rehabilitation and to determine when a patient can safely return to sport. [Figure 3](#)



Figure 3 Single leg squat functional test. (Color version of figure is available online.)

represents a single leg squat used in functional testing. In a recent systematic review, Webster and Hewett evaluated the validity of each functional test, finding limited evidence that passing functional testing led to reduction in subsequent knee injury upon return to sport. Of note, the authors found that while passing return to sport testing reduced risk of graft rupture, it also increased risk of contralateral ACL injury.³³ As demonstrated by Nagai et al, caution must be utilized when considering which functional tests to use.³⁴ When comparing isokinetic dynamometry testing, hop tests, and leg press, the authors found significantly higher LSI values in single and triple hop tests compared to isokinetic dynamometry testing and leg press, suggesting that hop tests may provide a falsely high LSI. However, in the evaluation of quadriceps strength, a positive association was appreciated between isokinetic dynamometry (the current “gold standard” for evaluating quadriceps muscular strength) and both hop distance and leg power during leg press, suggesting the clinical utility of all 3 tests in assessment of quadriceps strength.³⁴ Varying

combinations of these tests are commonplace and may provide useful feedback to patients, but they test the muscular envelope around the knee rather than the ACL graft, making them poor surrogates for assessing ACL healing, which does make full assessment of “ACL healing” difficult.

Post-Operative Bracing

Postoperative braces are routinely used following ACLR in the United States, though guidelines vary on how long to wear a brace locked in full extension and when to transition to a functional brace (commonly around 2 weeks after surgery).^{22,23} Despite common use, many studies have shown that postoperative braces do not provide clinical benefit.^{20,35} A systematic review by Lowe et al in 2017 found a lack of consensus on the efficacy of postoperative functional bracing 3–48 months following primary ACLR. Of the 15 included studies, 6 examined knee kinematics, each concluding that bracing is associated with improvement in arthrokinematics. The authors did not find a significant difference in knee function, joint laxity, range of motion, or subjective clinical outcomes between braced, and nonbraced groups.³⁶ A notable exception is the use of functional bracing in skiers. In a prospective cohort study, Sterett et al found that nonbraced skiers were almost 3 times as likely to experience re-injury than braced skiers at least 2 years after ACLR.³⁷

Neuromuscular Training

Neuromuscular training may be added to rehabilitation protocols to improve dynamic joint stability through development of movement patterns that allow for safe return to activities of daily living.³⁸ Supplementing standard rehabilitation protocols with neuromuscular control exercises (straight leg raise, single leg stance, lunges, step-up, heel slides, quadriceps isometric exercises) has been shown by Kaya et al to be beneficial in improving joint position sense and decreasing the strength index between operative and nonoperative knees at high angular speeds.³⁹ Neuromuscular training was also found to improve Cincinnati Knee Scores and Visual Analog Scores (VAS) for function at 6-month and 1-year follow-up compared to standard rehabilitation.^{40,41} A systematic review by Kruse et al that included 9 randomized clinical trials concluded that although neuromuscular training is likely not harmful, there is limited evidence that it provides long term benefit to patients or substantial improvements compared to standard protocols.⁴²

Electrostimulation and Cryotherapy

Electrostimulation and electromyographic feedback are less commonly used modalities in ACL rehabilitation. In a systematic review, van Melick et al found that the addition of electrostimulation to rehabilitation increases quadriceps strength up to 6 months following ACLR but likely does not affect effusion rates, pain scores, or range of motion.^{43–45}

Electromyographic feedback specifically in ACL rehabilitation has not been extensively studied. Wasielewski et al found that electromyographic feedback may decrease post-surgical pain and improve quadriceps strength, but the authors caution that evidence is limited to support this statement.⁴⁶ Christanell et al found that electromyographic feedback in ACL rehabilitation improves passive knee extension in the short term postoperative period (2 weeks) but that this measurement was not significantly different from the control group at 6 month follow-up.⁴⁷ Cryotherapy is often used following ACL reconstructive surgery, although a systematic review in 2014 found limited evidence on cryotherapy's effect on edema, knee function, blood loss, range of motion, or patient satisfaction. Nevertheless, cryotherapy was shown to decrease postoperative pain for up to 48 hours.⁴⁸

Blood Flow Restriction Therapy

Blood flow restriction training in an emerging trend in ACL rehabilitation, during which a pneumatic cuff or band is applied to the lower extremity to partially occlude venous blood flow while performing low-intensity resistance training (Fig. 4).⁴⁹ This technique induces anaerobic conditions that lead to upregulation of vascular endothelial growth factor and increased muscular strength.^{50–52} To date, the long-term outcomes of blood flow restriction training have not been extensively studied. However, in a recent systematic review, Lu et al found that recent clinical trials have demonstrated early postoperative blood flow restriction leading to improved subjective outcomes and increased muscle hypertrophy and strength compared to control groups at early follow-up (up to 16 weeks postoperatively).^{52–56} Though systematic reviews have shown good outcomes for blood flow restriction therapy following ACLR, many of the included studies are varied in type of tourniquet device used, ranging from wide, rigid, inelastic devices to narrow, elastic bands.^{52,53} Additional research is necessary to reach a consensus on blood flow restriction therapy protocols.

Biomechanics Analysis Outcomes

Despite the recommendation for the use of symmetry index during functional tests, there is growing literature showing that achieving symmetric distances during single leg hops does not ensure adequate biomechanical alignment of lower limbs.⁵⁷ Three-dimensional biokinetic analysis has emerged as a highly sensitive technology that can objectively evaluate joint function, be used as a surrogate measure of lower limb loads, and describe risk factors for reinjury or development of knee OA (Fig. 5).^{58,59}

In a systematic review evaluating the biomechanical outcomes of ACL reconstruction surgery, Moore et al (2020) found that the variables with most consistent improvement from pre to post surgery were joint position sense and static

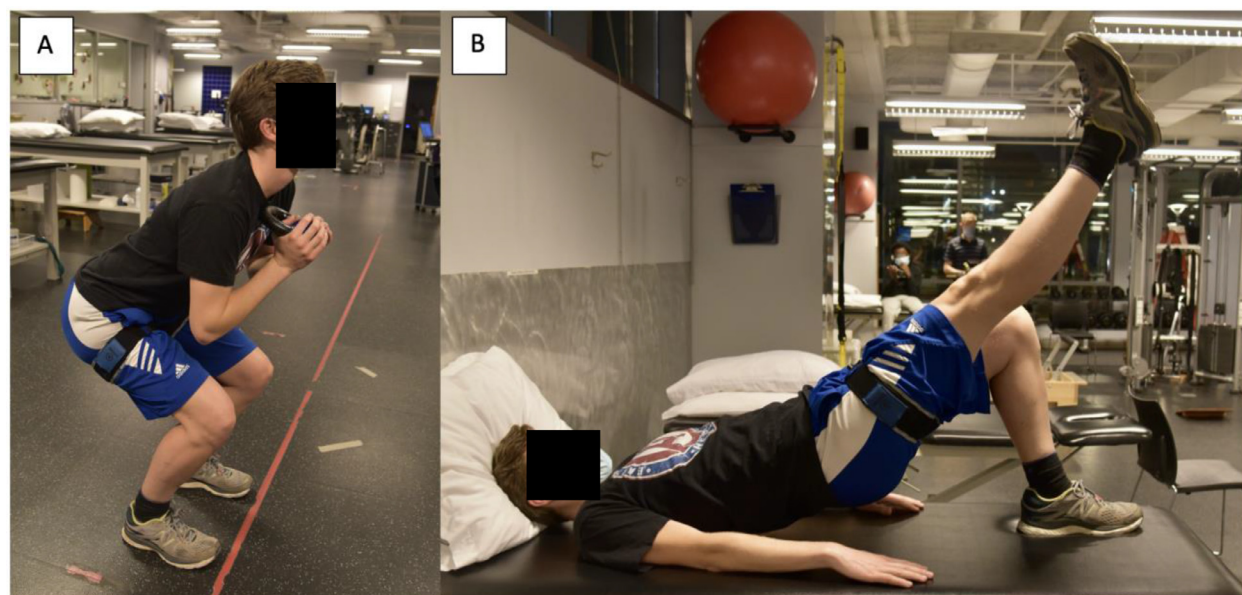


Figure 4 Blood flow restriction training during ACLR rehabilitation. (A) Weighted squat during blood flow restriction training. (B) Straight leg bridge. (Color version of figure is available online.)

balance. Both kinematic and kinetic improvement were inconsistent among articles. In general, active knee flexion range of motion during gait increased after surgery, suggesting a greater use of the operated limb.⁵⁸ Knee functional stability has shown to be task intensity dependent. At 24 and 48 weeks after ACL reconstruction, Moore et al. (2020) did not find improvements in tibial rotation angles and excursion during gait. However, they identified reduction in tibial anterior and rotational movements (an indicator of better knee stability) for pivoting and landing tasks. It seems that simple gait is not the adequate task to impose enough load to the knee to challenge rotational stability after ACL reconstruction and rehabilitation. This task dependence should also be considered when comparing function of ACLR subjects with the contralateral limb or non-injured individuals. While some authors describe ACLR subjects landing, running, and changing direction with lower internal knee and hip extension moments resulting in larger loads on the ACL, others have demonstrated ACLR subjects landing with greater, lower, or similar knee flexion compared to contralateral limb and non-injured individuals.^{57,59–62} Peak knee adduction seems to be reduced during double-leg landings, despite non conclusive results for single-leg landings.^{57,59} Ground reaction force measures are controversial since either a decrease or no difference have been reported to be independent of the task performed.^{57,59}

The contradictory results of the 3D kinematic motion studies are insightful to the fact that ACLR subjects assume different strategies to re-establish functionality, similar to what has already been shown for those with ACL injuries.⁶³ These kinematic changes may not be easily identifiable clinically. For that reason, pre-surgery and post-surgery periodical evaluations, besides being helpful to establish biomechanical outcomes, may adequately identify individual adaptations, allowing for implementation of a customized

physical rehabilitation approach, safe earlier return to activity, and prevention of further injury.

Return to Play

Similar to the inconsistency across rehabilitation protocols after ACLR, guidelines for return to play are quite variable. In a systematic review of Level I randomized controlled trials, Harris et al found inconsistency in both description and reporting of return to sport guidelines. Less than half of the studies used objective criteria in the decision to recommend return to sport. Of the 49 studies analyzed, 39% allowed running 3 months after surgery, 45% allowed sports with cutting and pivoting 6 months after surgery, and 51% had no restrictions 6 months after surgery. Some reported return to sport criteria were full functional rehabilitation and stability, full motion, full strength, strength >90% of contralateral knee, and lack of effusion.⁶⁴ Despite these inconsistencies, many of the metrics used to determine return to play are already measured in rehabilitation (single-leg hop, isokinetic strength, KT-1000 or KT-2000 arthrometer), highlighting the importance of physical therapy in allowing patients to resume sporting activities.⁶⁴ Although rehabilitation continues to play an important role in how physicians and patients measure progress, stricter criteria with guidance on return to sport timelines and objective data may be beneficial in prevention of further injury.

Pre-Injury Prevention

Because ACL injuries can be devastating to athletes, ACL injury prevention programs are widely utilized. In a meta-

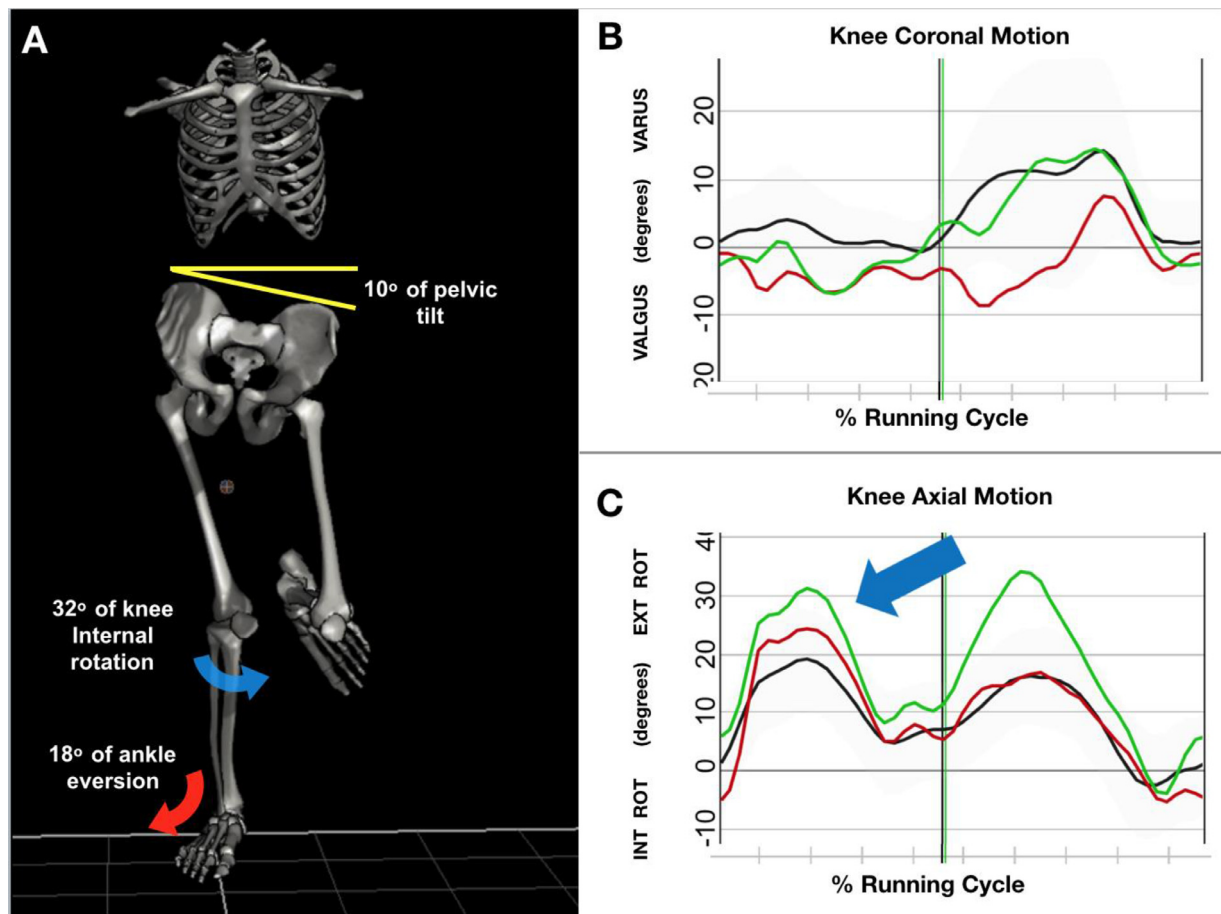


Figure 5 (A) Three-dimensional biokinetic analysis image in the coronal plane in the stance phase of running of a 35-year-old male recreational long-distance runner evidencing an excessive knee internal rotation peak (32°) influenced by ankle eversion (18°) and contralateral pelvic drop (10°); (B) Running cycle graph in the coronal plane demonstrating slight knee valgus; (C) Running cycle graph in the axial plane demonstrating excessive knee internal rotation peak. Blue arrows: excessive knee internal rotation peak; Red arrow: excessive ankle eversion; Yellow lines: excessive contralateral pelvic drop; Green line: right knee motion; Red line: left knee motion; Black line: expected knee motion (Image by Biocinetica Laboratório do Movimento Ltda; reprinted with permission from J Foot Ankle 2020;14(2):191-6). Three-dimensional biokinetic analysis image and corresponding running cycle graphs (Color version of the figure is available online.)

analysis including over 11,000 soccer players, Grimm et al found a significant decrease in the risk of knee injury for athletes participating in an ACL injury prevention program.⁶⁵ Similar outcomes have been described for handball, basketball, volleyball, and football.^{66–68} In order to prevent ACL injuries, one must have an understanding of the types of movements that put an athlete at risk of injury. Much of the focus of these programs is on landing from a jump as this has been cited as the mechanism of injury in up to 80% of ACL tears.⁶⁹ The following 4 theories have been validated with video analysis as being associated with an ACL injury when landing from a jump: ligament dominance (knee in valgus with an adducted and internally rotated femur), quadriceps dominance (knee extended with a high level of quadriceps activation compared to hamstrings activation), trunk dominance (trunk control deficits), and leg dominance (side-to-side leg asymmetries).^{69–72}

In a systematic review of the literature regarding landing biomechanics following ACL injury, Lopes et al found that

participation in an injury prevention program led to increased knee and hip flexion during landing and decreased peak knee abduction during landing, suggesting that these programs may be successful through alteration of an athlete's landing patterns.⁷³ Through an assessment of power, balance, strength, and landing force in female athletes that underwent ACL training programs focused on either plyometrics or dynamic stabilization and balance training, Myer et al found that although there were similar outcomes of neuromuscular power, body sway, and dynamic strength between the programs, there was a significant difference in impact force when landing from a single leg jump.⁷⁴ Athletes in the plyometrics group increased their impact force by 7.6%, while those in the balance group decreased by 7% on the dominant side, highlighting the importance of focusing on balance and dynamic stabilization to improve jump landing force attenuation in ACL injury prevention programs.⁷⁴

Aside from modification of jump landing biomechanics, ACL injury prevention programs focus on neuromuscular

control and strength training. Through improvement of neuromuscular control, athletes are better able to control frontal plane motion during landing and cutting movements and increase strength, which can prevent discrepancies between recruitment of the knee extensors and flexors and between legs during sporting activities.⁷⁵ Attention should also be paid to trunk control during ACL injury prevention as core strength deficits can result in excessive frontal plane trunk displacements and higher knee abduction loads, increasing the risk of injury.^{75,76}

Conclusions

Although ACL injury prevention programs that focus on modification of jump landing biomechanics, neuromuscular control, and strength training have demonstrated efficacy in a variety of sporting activities, proper rehabilitation must be used if injury does occur. Regardless of whether rehabilitation is used for operative or nonoperative ACL-injured patients, recent studies have demonstrated improved clinical outcomes with both in-person and at-home rehabilitation protocols. Certain specific modalities, such as blood flow restriction training and neuromuscular training, may provide great benefit in rehabilitation, while others (such as bracing) may not be necessary. Biomechanical analysis may also allow for the development of individualized rehabilitation protocols. The lack of consensus amongst rehabilitation protocols poses the question of whether standardization may lead to improved clinical outcomes and safer, faster return to play.

Disclosure

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